

Nutanix Hackathon IIT Guwahati

Project Areas and Guidelines

NUTANIX

Hackathon Project Areas (Part 1)

Sl. No	Project Area	Detailed Information
1	The Zero-Trust Intelligent Gateway (Envoy)	Modern distributed architectures require robust edge and service proxies to manage traffic, enforce security, and provide deep observability. Bring in your own ideas in enhancing Envoy Proxy which acts as an intelligent, zero-trust edge gateway that satisfies real-world production constraints across scale, security, and performance.
2	Develop mechanism to keep the files in sync across two machines	There is need to keep the files in multiple machines of a cluster in sync. Bring-in your ideas and build solution for this case. The solution should be scalable and performant and be able to deal with 100s of machines.
3	Persistent Message Broker (NATS)	Bring in your ideas in building Message Broker with Pub-Sub model. The delivery guarantees include as per client need: Max Once, At least once. Provide mechanism to retain message for certain duration or based on certain capability.

Hackathon Project Areas (Part 2)

Sl. No	Project Area	Detailed Information
4	Workflow execution Engine	Bring in your ideas in capturing the workflow input via a json payload and provide ability to execute it. The workflow execution should have ability to handle shell script, REST API Calls. Add extensibility as per your views.
5	Cybersecurity	Bring in your ideas to build a cybersecurity system that intercepts and defeats modern malware/ransomware before the damage spreads. The challenge is creating a real-time detection engine to instantly spot suspicious patterns, block the attack with minimal performance overhead, and include a smart way to quickly recover safe data.
6	AI Assistant for Engineers	Rethink use of AI as coding assistance; the helper that enables faster delivery and better quality of projects. Leverage power of AI for for engineers to quickly understand high level architecture picture of projects, help visualize individual component design, help code reviews, test case generation and verification, UI visualization, performance modeling etc.. A tool that goes beyond just coding.
7	Post Quantum Cryptography Proxy	Build a post-quantum reverse proxy that terminates TLS 1.3 traffic using NIST-approved hybrid key exchange algorithms (ML-KEM/Kyber) and forwards unencrypted traffic to an upstream web application. Implement an AI-driven analytics dashboard that intercepts, benchmarks, and detects traffic anomaly patterns stemming from post-quantum cryptographic overhead.
8	Surprise Us!	Bring In your own ideas on any of the topic of your interest and showcase your implementation.

Evaluation Criteria

- Innovation & creativity
- End to End Execution (Completeness) - Working State of the Project
- Practical impact and usefulness
- Presentation & Articulation (Demo)

Refined Evaluation Criteria

Refined Criteria	Refined Definition
Innovation & Creativity	Originality of the idea and the creativity demonstrated in exploring or addressing the problem.
Overall Completeness	Quality and completeness of the solution achieved within the given time, reflecting the team's effort and commitment rather than production readiness.
Presentation & Articulation	Clarity and effectiveness in communicating your idea, approach, key outcomes, and learnings, including how well the team conveys their thought process and insights.
Learning & Skill Application	Effective use of concepts, tools, or technologies learned during the internship or hackathon, with evidence of learning and experimentation.

Guidelines

Hackathon Schedule

Saturday to Sunday (Sep 5, 2025 - Sep 6, 2025)

Team Composition

Each team may consist of up to **six members**.

All code must be original; AI-generated code or code copied from external sources is prohibited.

(Code will be verified for originality.)

Submission Requirements

- Prepare a recorded **DEMO VIDEO (5 mins or less) AND a POWERPOINT PRESENTATION** and submit it here [Nutanix Hackathon Submission](#) by **Sep 7, 2026 by 11 AM. No submissions will be accepted post 12 PM.**
- **Selected finalists will be invited to present a live virtual demo to the Nutanix panel.**

Winner Announcements

Winners will be declared at **IIT Guwahati on Sep 21, 2026**

Awards!!

We're thrilled to announce exciting cash prizes to recognize and reward the innovative solutions developed during our upcoming Hackathon.

CATEGORY	PRIZE (Per team)
Winners	INR 30,000
First Runner-Up	INR 20,000
Second Runner-Up	INR 15,000



Thank You